

UNIT-1

Chapter-1 introduction to databases

Definition

A **database** is an organized collection of related data that is stored electronically and can be easily accessed, managed, updated, and retrieved.

Example

A **Student Database** stores information such as:

Student_ID	Name	Course	Age
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101	Narendra	BCA	20
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102	Rahul	BCA	19
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Characteristics of a Database

- Stores related data in an organized manner.
- Reduces data redundancy (duplication).
- Allows easy data retrieval and updating.
- Supports multiple users.
- Ensures data security and integrity.

Advantages of a Database

1. Easy storage of large amounts of data.
2. Fast retrieval of information.
3. Reduces data duplication.
4. Improves data security.
5. Provides backup and recovery.
6. Supports data sharing among users.

Exam Definition (2 Marks)

A **database** is an organized collection of related data that is stored electronically and can be easily accessed, managed, and updated.

Exam Question (5 Marks)

What is a Database? Explain with an example.

Answer:

A database is an organized collection of related data that is stored electronically for easy access, management, and retrieval. It helps reduce data redundancy, improves data security, and allows multiple users to access the data efficiently.

Example: A college student database containing student ID, name, course, and age.

Database System**Definition**

A **Database System** is a combination of the **database**, the **Database Management System (DBMS)**, and the **users/applications** that interact with the database to store, retrieve, update, and manage data efficiently.

Simple Formula

Database System = Database + DBMS + Users/Applications

Components of a Database System

1. **Database** – Stores the data.
2. **DBMS** – Software that manages the database.
3. **Users** – People who access and use the data.
4. **Application Programs** – Software used to interact with the database.
5. **Hardware** – Computers and storage devices.
6. **Procedures** – Rules and instructions for using the database.

Example**A College Database System:**

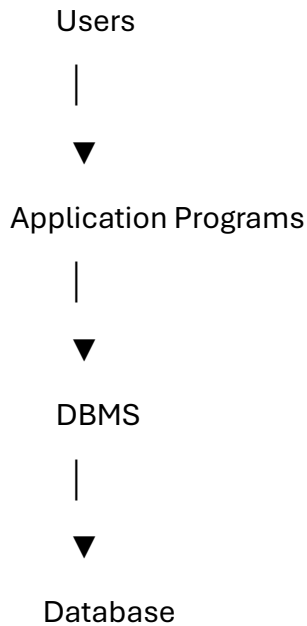
- **Database:** Student records, faculty details, and course information.
- **DBMS:** MySQL or Oracle.
- **Users:** Students, teachers, and administrators.
- **Application:** College management software.

Advantages

- Easy data storage and retrieval.
- Reduces data redundancy.
- Provides data security.
- Supports multiple users.

- Backup and recovery of data.
- Maintains data consistency.

Diagram



Exam Definition (2 Marks)

A Database System is a combination of a database, DBMS, users, and application programs that work together to store, manage, and retrieve data efficiently.

Exam Question (5 Marks)

What is a Database System? Explain its components.

Answer:

A Database System is a collection of a database, DBMS, users, application programs, hardware, and procedures that work together to store, organize, manage, and retrieve data efficiently. It provides secure, consistent, and efficient access to data while supporting multiple users.

DBMS (Database Management System)

Definition

A Database Management System (DBMS) is software that is used to create, store, organize, retrieve, update, and manage data in a database efficiently.

A DBMS acts as an interface between the **user** and the **database**, allowing users to access and manipulate data without needing to know how it is physically stored.

Examples of DBMS

- MySQL
- Oracle

- Microsoft SQL Server
- PostgreSQL
- SQLite
- IBM Db2

Functions of DBMS

1. Data storage and retrieval
2. Data insertion, deletion, and updating
3. Data security and authorization
4. Backup and recovery
5. Data sharing among multiple users
6. Maintaining data integrity and consistency

Advantages of DBMS

- Reduces data redundancy (duplicate data)
- Improves data security
- Supports data sharing
- Provides backup and recovery
- Ensures data integrity
- Allows multiple users to access data simultaneously

Disadvantages of DBMS

- High installation and maintenance cost
- Requires skilled professionals
- Uses more memory and storage
- System failure may affect all users

Diagram

User

|



DBMS

|



Database

Exam Definition (2 Marks)

A Database Management System (DBMS) is software that enables users to create, store, retrieve, update, and manage data in a database efficiently.

Importance of Database

Definition

A **database** is important because it helps in storing, organizing, managing, and retrieving data efficiently. It provides a systematic way to handle large amounts of information.

Importance of Database

1. Organized Data Storage

- Stores data in a structured and systematic manner.
- Makes data easy to manage.

2. Easy Data Retrieval

- Data can be searched and accessed quickly whenever required.

3. Reduces Data Redundancy

- Avoids storing duplicate copies of the same data.
- Saves storage space.

4. Improves Data Security

- Protects data from unauthorized access.
- Allows only authorized users to access the database.

5. Maintains Data Integrity

- Ensures that the data is accurate, consistent, and reliable.

6. Supports Data Sharing

- Multiple users can access and share the same database simultaneously.

7. Backup and Recovery

- Provides facilities to back up data and recover it in case of system failure.

8. Better Decision Making

- Helps organizations make accurate and informed decisions using reliable data.

9. Saves Time

- Enables fast insertion, updating, deletion, and retrieval of data.

10. Efficient Data Management

- Makes it easier to maintain and manage large volumes of data.
-

Exam Definition

Data Manipulation is the function of DBMS that allows users to insert, update, delete, and retrieve data from the database.

3. Data Security and Integrity

Meaning

This function protects the database from unauthorized access and ensures that the data remains accurate, consistent, and reliable.

Example

- Only authorized users can access confidential records.
- Preventing invalid or duplicate data entries.

Exam Definition

Data Security and Integrity is the function of DBMS that protects data from unauthorized access and maintains its accuracy and consistency.

4. Data Dictionary Maintenance

Meaning

A **Data Dictionary** is a collection of information (metadata) about the database, such as table names, column names, data types, constraints, and relationships.

DBMS automatically maintains and updates this information.

Example

- Keeping details of the **Student** table, its columns, and their data types.

Exam Definition

Data Dictionary Maintenance is the function of DBMS that stores and manages metadata (information about the database structure).

5. Performance

Meaning

The **Performance** function of DBMS ensures that database operations are performed quickly and efficiently, even when handling large amounts of data.

Example

- Retrieving student records in a few seconds.
- Using indexes to speed up searches.

Exam Definition

Performance is the function of DBMS that optimizes database operations to provide fast and efficient data access.

Summary Table

Function	Meaning
Data Definition	Defines the structure of the database (tables, columns, data types).
Data Manipulation	Inserts, updates, deletes, and retrieves data.
Data Security and Integrity	Protects data and maintains accuracy and consistency.
Data Dictionary Maintenance	Maintains metadata about the database structure.
Performance	Improves the speed and efficiency of database operations.

Database Applications

Definition

Database applications are software systems that use databases to store, manage, retrieve, and process data efficiently in different fields.

Applications of Database

1. Banking

- Stores customer account details.
- Manages deposits, withdrawals, and transactions.
- Maintains loan and ATM records.

2. Education

- Stores student, faculty, and course information.
- Manages attendance, examination, and result records.

3. Hospital

- Maintains patient records.
- Stores doctor information and appointment details.
- Manages medical history and billing.

4. Railway Reservation

- Stores train schedules and passenger details.
- Handles ticket booking, cancellation, and seat availability.

5. E-Commerce

- Stores customer, product, and order information.
- Manages online shopping, payments, and delivery.

6. Library Management

- Maintains book details and member records.
- Tracks book issue, return, and fine details.

7. Telecommunications

- Stores customer information.
- Manages call records, recharge, and billing.

8. Government Organizations

- Maintains citizen records.
- Stores Aadhaar, voter ID, passport, and tax information.

9. Airline Reservation

- Stores flight schedules and passenger details.
- Handles ticket booking and cancellations.

10. Business and Finance

- Manages employee, customer, inventory, payroll, and sales records.

Advantages of Database Applications

- Fast data retrieval
- Reduced data redundancy
- Improved data security
- Easy data sharing
- Accurate and reliable information
- Backup and recovery

• **Advantages of DBMS**

• **1. Reduces Data Redundancy**

- **Meaning:**
DBMS avoids storing the same data multiple times.
 - **Example:**
A student's address is stored only once in the database instead of being repeated in different files.
-

• **2. Improves Data Security**

- **Meaning:**
Only authorized users can access or modify the data.
 - **Example:**
Only the bank manager can update customer account details, while customers can only view their own accounts.
-

• **3. Maintains Data Integrity**

- **Meaning:**
DBMS ensures that data remains accurate, valid, and consistent.
 - **Example:**
A student's age cannot be entered as -5 because the database validates the data.
-

• **4. Data Sharing**

- **Meaning:**
Multiple users can access the same database at the same time.
 - **Example:**
In a college, teachers, students, and the principal can access the student database simultaneously.
-

• **5. Backup and Recovery**

- **Meaning:**
DBMS creates backups and restores data if it is lost.
 - **Example:**
If a hospital computer crashes, patient records can be recovered from the backup.
-

• **6. Data Consistency**

- **Meaning:**
The same updated data is available to all users.

- **Example:**
If a student's phone number is updated once, every department sees the new phone number.
-

• **7. Easy Data Retrieval**

- **Meaning:**
Data can be searched and retrieved quickly.
 - **Example:**
A librarian can search for a book using its title or ISBN within seconds.
-

• **8. Better Data Management**

- **Meaning:**
DBMS stores data in an organized and systematic manner.
 - **Example:**
An employee database keeps records such as employee ID, name, department, and salary in separate tables.
-

• **9. Concurrency Control**

- **Meaning:**
Many users can access the database simultaneously without causing data conflicts.
 - **Example:**
Several passengers can book train tickets at the same time without receiving the same seat.
-

• **10. Faster Decision Making**

- **Meaning:**
DBMS provides accurate and up-to-date information to help in decision-making.
 - **Example:**
A shop owner checks monthly sales reports from the database to decide which products should be restocked.
-

• **Disadvantages of DBMS (With Examples)**

• **Definition**

- A **Database Management System (DBMS)** has many advantages, but it also has some disadvantages such as high cost, complexity, and the need for skilled professionals.
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• **1. High Cost**

- **Meaning:**
Installing and maintaining a DBMS is expensive.
 - **Example:**
A company has to purchase licensed software like Oracle and pay for maintenance.
-

• **2. Complexity**

- **Meaning:**
DBMS software is complex to install, configure, and manage.
 - **Example:**
A beginner may find it difficult to create and manage a database without proper training.
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• **3. Requires Skilled Professionals**

- **Meaning:**
A trained Database Administrator (DBA) is needed to manage the database.
 - **Example:**
A bank hires experienced DBAs to maintain customer databases.
-

• **4. Large Storage Requirement**

- **Meaning:**
DBMS software requires more memory and disk space.
 - **Example:**
A hospital database storing millions of patient records needs large storage devices.
-

• **5. Performance Overhead**

- **Meaning:**
DBMS may be slower for small applications because it performs many security and management tasks.
 - **Example:**
A simple student record program may run faster using a basic file system than a full DBMS.
-

• **6. Single Point of Failure**

- **Meaning:**
If the DBMS server fails, users may not be able to access the database.
 - **Example:**
If a bank's database server crashes, customers cannot perform online transactions until it is restored.
-

• **7. Regular Maintenance Required**

- **Meaning:**
Databases require regular backups, updates, and performance tuning.
 - **Example:**
A university database must be backed up regularly to prevent data loss.
-

• **8. Security Risks**

- **Meaning:**
If the database is hacked or security is weak, a large amount of data can be exposed.
- **Example:**
A cyberattack on an online shopping website may leak customer information.

Comparison Between Traditional File-Based Approach and Database Approach

Definition

- **Traditional File-Based Approach:** Data is stored in separate files for each application, making data management difficult.
 - **Database Approach:** Data is stored in a centralized database and managed by a DBMS.
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Differences

Traditional File-Based Approach

Data is stored in separate files.
High data redundancy (duplicate data).
Data inconsistency is common.
Security is limited.
Data sharing is difficult.
Backup and recovery are difficult.
Less data integrity.
Programs depend on file structure.
Difficult to maintain.
Suitable for small applications.

Database Approach

Data is stored in a centralized database.
Low data redundancy.
Data consistency is maintained.
Strong security is provided by the DBMS.
Data sharing is easy among multiple users.
Backup and recovery are easy.
High data integrity.
Programs are independent of data structure (data independence).
Easy to maintain and update.
Suitable for small as well as large applications.

Example

Traditional File-Based Approach

A college stores:

- Student details in one file.
- Fee details in another file.
- Exam details in a third file.

The same student information is repeated in all three files.

Database Approach

A college stores all student information in one database. Different departments access the same data without duplication.

A bank employs a DBA to manage customer account databases.

2. Database Designers

Meaning

Database designers design the structure of the database according to user requirements.

Responsibilities

- Design database tables.
- Define relationships between tables.
- Create the database schema.
- Ensure efficient database design.

Example

A designer creates a **Student** database with tables for Students, Courses, and Faculty.

3. System Analysts and Application Programmers

Meaning

They analyze user requirements and develop application programs that interact with the database.

Responsibilities

- Gather user requirements.
- Develop database applications.
- Write SQL queries and programs.
- Test and maintain applications.

Example

A programmer develops a **Library Management System** using MySQL.

4. End Users

Meaning

End users are people who use database applications to perform their daily tasks.

Types of End Users

- **Casual Users:** Access the database occasionally. (*Example: Manager viewing reports.*)
- **Naive (Parametric) Users:** Use predefined forms. (*Example: Bank teller entering customer transactions.*)
- **Sophisticated Users:** Write SQL queries and perform complex analysis. (*Example: Data analyst.*)
- **Standalone Users:** Use personal database applications. (*Example: A shop owner using MS Access.*)

backup tool or report generation software.

3. Operators and Maintenance Personnel

Meaning

They operate and maintain the hardware and software used by the DBMS.

Responsibilities

- Install and configure the DBMS.
- Monitor system performance.

- Perform backups and updates.
- Resolve hardware and software issues.

Example

A system operator maintaining the database server in a company.

DBA (Database Administrator)

Definition

A **Database Administrator (DBA)** is a person who is responsible for managing, controlling, maintaining, and securing the database system. The DBA ensures that the database operates efficiently, securely, and reliably.

Roles and Responsibilities of DBA

1. Database Installation and Configuration

Meaning

The DBA installs the DBMS software and configures it according to the organization's requirements.

Example: Installing MySQL or Oracle Database on a company server.

2. Database Design and Creation

Meaning

The DBA creates databases, tables, indexes, and other database objects.

Example: Creating a **Student** database with Student, Course, and Faculty tables.

3. User Management

Meaning

The DBA creates user accounts and grants appropriate access permissions.

Example: Giving teachers permission to update student marks, while students can only view their results.

4. Data Security

Meaning

The DBA protects the database from unauthorized access and cyber threats.

Example: Setting passwords and access controls for bank customer records.

5. Backup and Recovery

Meaning

The DBA regularly backs up the database and restores data if it is lost or damaged.

Example: Recovering a hospital database after a system crash.

6. Performance Monitoring and Tuning

Meaning

The DBA monitors database performance and optimizes it for faster operation.

Example: Creating indexes to speed up student record searches.

7. Data Integrity

Meaning

The DBA ensures that the data stored in the database is accurate, consistent, and reliable.

Example: Preventing duplicate student registration numbers.

8. Database Maintenance

Meaning

The DBA performs regular updates, maintenance, and troubleshooting to keep the database running smoothly.

Example: Updating the DBMS software to the latest version.

9. Concurrency Control

Meaning

The DBA manages multiple users accessing the database simultaneously without conflicts.

Example: Many customers using online banking at the same time.

10. Disaster Recovery Planning

Meaning

The DBA prepares plans to recover the database in case of hardware failure, cyberattacks, or natural disasters.

Example: Restoring company data from backup after a server failure.

Short Note on DBA (Database Administrator)

Definition

A Database Administrator (DBA) is a person responsible for managing, maintaining, securing, and controlling the database system. The DBA ensures that the database operates efficiently, securely, and reliably.

Responsibilities of DBA

- 1. Installs and configures the DBMS.**
- 2. Creates and maintains databases.**
- 3. Manages user accounts and access permissions.**
- 4. Ensures data security and privacy.**
- 5. Performs backup and recovery of data.**
- 6. Maintains data integrity and consistency.**
- 7. Monitors and improves database performance.**
- 8. Updates and maintains the database system.**

Example

In a bank, the DBA manages customer account databases, performs regular backups, controls user access, and protects customer information from unauthorized access.

Advantages of Having a DBA

- Ensures data security.**
- Prevents data loss through backups.**
- Improves database performance.**
- Maintains accurate and consistent data.**
- Supports smooth database operations**

Chapter-2 data models

Data Model

A data model is a collection of concepts used to describe the structure of data, relationships among data, and rules for storing and manipulating data in a database.

In simple words, a data model tells us how data is organized, how different data items are related, and how we can work with the data.

Components of Data Models

A data model mainly consists of three components:

1. Structural Part

The structural part describes the structure of the database. It defines:

- What data is stored
- How data is organized
- Relationships between different data items
- Entities, attributes, and tables

Example:

Consider a college database:

- Student $\rightarrow$ Student_ID, Name, Course
- Course $\rightarrow$ Course_ID, Course_Name

Here, the structure shows that a student belongs to a particular course.

Example:

STUDENT(Student_ID, Name, Course_ID)

COURSE(Course_ID, Course_Name)

This represents how the data is structured and related.

2. Manipulative Part

The manipulative part defines the operations that can be performed on the data.

These operations include:

- Insert new data
- Update existing data
- Delete data

- Retrieve or search data

Example:

In SQL, we can perform these operations:

INSERT INTO Student VALUES (101, 'Rahul', 10);

Adds a new student.

SELECT * FROM Student;

Retrieves student information.

UPDATE Student

SET Name = 'Ravi'

WHERE Student_ID = 101;

Updates student information.

DELETE FROM Student

WHERE Student_ID = 101;

Deletes a student record.

Thus, the manipulative part explains how we can access and modify the data.

3. Set of Integrity Rules

Integrity rules are rules that ensure the accuracy, consistency, and validity of data in a database.

They prevent invalid or incorrect data from being entered.

Common integrity rules include:

a) Entity Integrity

A primary key cannot contain NULL values.

Example:

Student

Student_ID Name

101 Rahul

Student_ID is the primary key, so every student must have a valid **Student_ID**.

b) Referential Integrity

A foreign key must refer to an existing primary key in another table.

Example:

If **Course_ID = 10** does not exist in the **Course** table, a student should not be assigned **Course_ID = 10**.

c) Domain Integrity

A column should contain values from its defined domain or valid range.

Example:

If **Age** is defined as an integer, values such as "ABC" should not be accepted.

Types of Data Models

A Data Model is a collection of concepts used to describe how data is organized, stored, and related in a database.

The four main types of data models are:

1. Hierarchical Data Model
2. Network Data Model
3. Relational Data Model
4. Object-Oriented Data Model

1. Hierarchical Data Model

Definition

The Hierarchical Data Model organizes data in a tree-like structure, where each parent record can have many child records, but each child has only one parent.

It follows a one-to-many (1:M) relationship.

Structure

College

|

| |

Department Department

|

| |

Student Student

Characteristics

- **Tree structure.**
- **One parent can have many children.**
- **Each child has only one parent.**
- **Supports one-to-many relationships.**
- **Fast data retrieval when the hierarchy is known.**

Advantages

1. **Simple and easy to understand.**
2. **Fast data access.**
3. **High data security.**
4. **Suitable for hierarchical data.**
5. **Efficient for one-to-many relationships.**

Disadvantages

1. **Does not support many-to-many relationships.**
2. **Difficult to modify the structure.**
3. **Data redundancy may occur.**
4. **Complex for large databases.**

Example

Organization Chart

Company

→ **Department**

→ **Employee**

2. Network Data Model

Definition

The Network Data Model organizes data as a network where one child record can have multiple parent records.

It supports many-to-many (M:N) relationships.

Structure

Student

/ \

/ \

Course1 Course2

\ /

\ /

Teacher

Characteristics

- **Graph-like structure.**
- **Supports many-to-many relationships.**
- **Records are connected using pointers.**
- **More flexible than the hierarchical model.**

Advantages

1. **Supports many-to-many relationships.**
2. **Fast data access.**
3. **Less data redundancy.**
4. **Flexible data structure.**
5. **Better performance for complex relationships.**

Disadvantages

1. **Complex structure.**
2. **Difficult to design.**
3. **Requires technical knowledge.**
4. **Database maintenance is difficult.**

Example

A student can enroll in many courses, and a course can have many students.

3. Relational Data Model

Definition

The Relational Data Model stores data in the form of tables (relations) consisting of rows and columns.

It is the most widely used data model.

Example

Student Table

Student_ID	Name	Course
101	Rahul	BCA
102	Narendra	BCA

Course Table

Course_ID	Course_Name
C101	DBMS
C102	Java

Characteristics

- **Data stored in tables.**
- **Rows represent records.**
- **Columns represent attributes.**
- **Uses primary and foreign keys.**
- **Supports SQL.**
- **Reduces data redundancy.**

Advantages

- 1. Easy to understand.**
- 2. Simple to maintain.**
- 3. Reduces data redundancy.**
- 4. High data integrity.**
- 5. Supports multiple users.**
- 6. Flexible database design.**
- 7. SQL makes querying easy.**
- 8. High security.**

Disadvantages

- 1. Performance may decrease with very large databases.**
- 2. Complex joins can be time-consuming.**
- 3. Requires more memory.**
- 4. Database design requires expertise.**

Example

College Management System

- **Student Table**
- **Faculty Table**
- **Course Table**

4. Object-Oriented Data Model

Definition

The Object-Oriented Data Model (OODM) stores data as objects, similar to object-oriented programming languages such as Java and C++.

An object contains both data (attributes) and methods (functions).

Example

Student Object

Student

Student_ID

Name

Course

Methods:

Display()

Update()

Characteristics

- Stores data as objects.
- Supports classes and objects.
- Supports inheritance.
- Supports encapsulation.
- Supports polymorphism.
- Suitable for multimedia applications.

Advantages

1. Supports complex data.
2. Easy integration with object-oriented programming languages.
3. Reduces code duplication.
4. Reusable objects.
5. Supports multimedia and graphics.
6. High flexibility.

Disadvantages

1. Complex to design.
2. Expensive implementation.
3. Less popular than the relational model.
4. Higher memory usage.
5. Learning curve is difficult.

Example

A multimedia database storing:

- Images
- Videos
- Audio
- Documents

• Importance of Data Models

• Definition

- A **Data Model** is a blueprint that defines how data is organized, stored, and related in a database. It helps in designing an efficient and well-structured database.

• Importance of Data Models

• 1. Helps in Understanding Data Requirements

• Meaning

- A data model helps designers and users understand what data is needed and how different data items are related.

• Example:

Before designing a **College Database**, the data model identifies entities such as **Student**, **Course**, and **Faculty**.

• 2. Acts as a Communication Tool

• Meaning

- A data model provides a common language for users, database designers, and developers to discuss the database design.

• Example:

A customer and a database designer use an **ER diagram** to discuss the required database.

• 3. Guides Database Design

• Meaning

- A data model serves as a blueprint for designing the database structure, including tables, relationships, and constraints.

• Example:

The **Relational Data Model** helps create tables such as **Student**, **Course**, and **Department**.

• 4. Helps in Future Expansion

• Meaning

- A well-designed data model makes it easy to add new tables or features without affecting existing data.
 - **Example:**
A college database can later include **Hostel** and **Library** tables without redesigning the entire database.
-

• **5. Improves Data Sharing Across Departments**

- **Meaning**
A data model allows different departments to access and share the same data efficiently.
 - **Example:**
The **Admission**, **Accounts**, and **Examination** departments use the same student database.
-

• **6. Supports Application Development**

- **Meaning**
A data model provides a clear database structure, making it easier for programmers to develop applications.
 - **Example:**
A programmer develops a **Library Management System** based on the designed database model.
-

• **7. Ensures Better Security**

- **Meaning**
A data model helps define user roles and access permissions to protect sensitive data.
 - **Example:**
Only the **Accounts Department** can update fee details, while students can only view them.
-

• **8. Improves Query Performance**

- **Meaning**
A good data model organizes data efficiently, making database queries faster.
 - **Example:**
A student's result can be retrieved quickly using a **Student ID**.
-

• **9. Reduces Development Cost and Time**

- **Meaning**
A proper data model reduces design errors, saving time and development costs.
- **Example:**
A company spends less time fixing database issues because the database was designed correctly from the beginning.